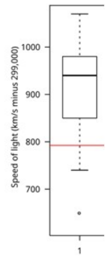
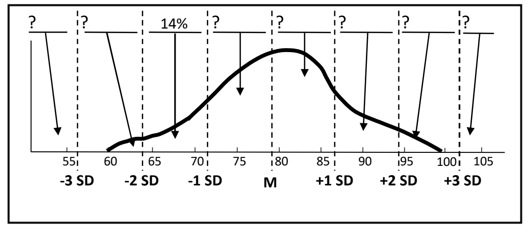
Agenda

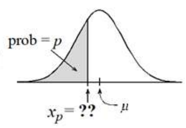


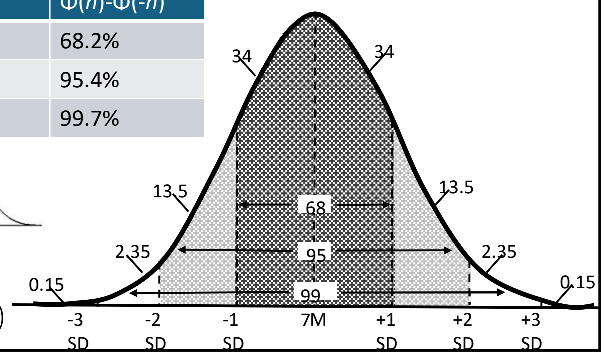
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



- Last time
  - Percentile and box-plots
  - Normal distribution
    - Transformations, Leibniz rule
      - z-score and Mahalanobis distance
      - Proportions
    - Inverse-probability or p-quantile
- Today
  - More on normal distribution (why?)

n	$\Phi(n)$-$\Phi(-n)$
1	68.2%
2	95.4%
3	99.7%





$$X \sim N(\mu, \sigma^2),\ Y = aX + b,\ Y \sim N(a\mu + b, a^2\sigma^2)$$

1

Why normal?

- So far, we have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - If each of the r.v. are identical
    - $E[Y] = nE[X_i]$
    - $Var[Y] = nVar[X_i]$
  - What about the distribution of Y?
    - Binomial: $X + Y \sim \text{Bin}(n + m, p)$

2

Sums of i.i.normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
- What is the distribution of Y?
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
  - Proof via Moment Generating Functions
  - First, the MGF of a unit normal

3

MGF of standard normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$E[e^{tZ}] = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2-2tx)/2} dx$$

$$= e^{-t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx$$

$$= e^{-t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy$$

$$= e^{-t^2/2}$$

4

MGF of Normal

- If Y = aX + b

$$\phi_Y(t) = e^{tb}\,\phi_X(at)$$

$$E[e^{tX}] = E[e^{t\mu + t\sigma Z}] = E[e^{t\mu}e^{t\sigma Z}] = e^{t\mu}E[e^{t\sigma Z}] = e^{t\mu}e^{-(\sigma t)^2/2} = e^{\mu t - \sigma^2 t^2/2}$$

5

Sums of i.i.normal distribution

- If X and Y are independent
- Aligning the MGF to the MGF in the previous slide we see
  - $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$

$$\phi_{X+Y}(t) = \phi_X(t)\,\phi_Y(t)$$

$$E\left[e^{t\sum_{i=1}^{n} X_i}\right] = \prod_{i=1}^{n} e^{\mu_i t + \sigma_i^2 t^2/2}$$

$$= e^{\mu t + \sigma^2 t^2/2}$$

6

Example

- Rain in Mumbai
  - Mean annual rainfall = 231 cm, standard deviation 56 cm.
  - Find the probability that rain in 2026 falls short of 2025 by 50 cm
- If we did not know the rainfall in 2025
  - Let $X_1, X_2 \sim \mathcal{N}(231, 56^2)$
  - Z = $X_1 - X_2 \sim \mathcal{N}(0, 2(56^2))$
  - $P\left(Z > 50/(56\sqrt{2})\right) = P(Z > 0.63) \approx .0.26$

Assumes r.v. X_1, X_2 are normal

Why normal?

- We have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - If each of the r.v. are identical
    - $E[Y] = nE[X_i]$
    - Var[Y] = nVar[X_i]
    - So the standard deviation of Y is related by the $\sqrt{n}$ factor
  - If each r.v. is also normal, then the distribution of Y is also normal
  - *What if the r.v. comes from identical, but from a non-normal distribution?*

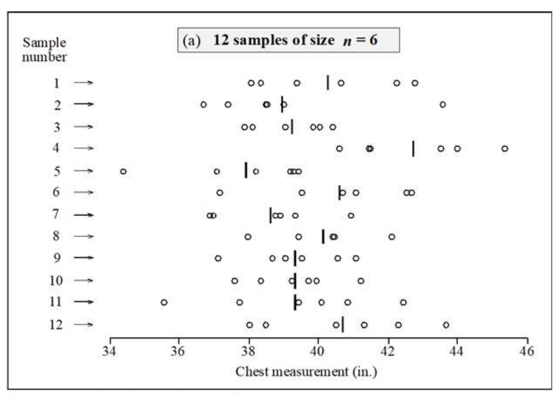
Why normal?

- We have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - What if the r.v. comes from a different distribution?
- Central limit theorem
  - For large n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
- Let's understand this theorem, and instead of looking at the sum, look at the expression $\frac{y}{n}$, i.e., the sample mean (for convenience)

10

Example

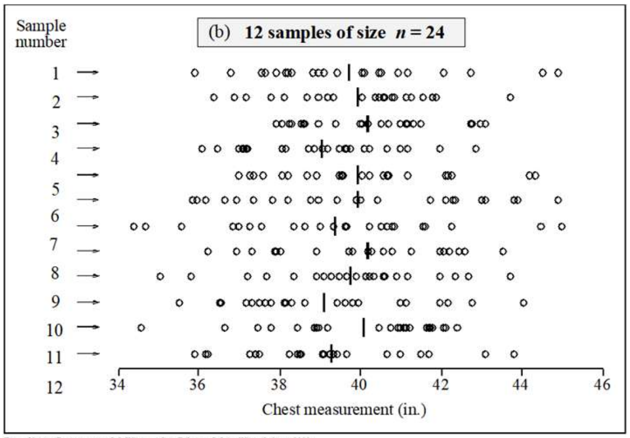
- Twelve samples of size 6 from 5768 soldiers
- Mean is shown with the vertical bar
- Observation: The sample mean is not the same
  - The sample mean is a random variable
  - We will use the sample mean as a proxy for the sums of random variables from an unknown distribution



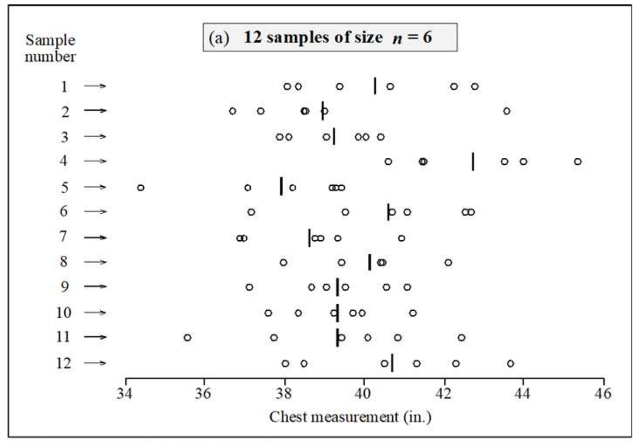
From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 1999.

11

Example



From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 2000.

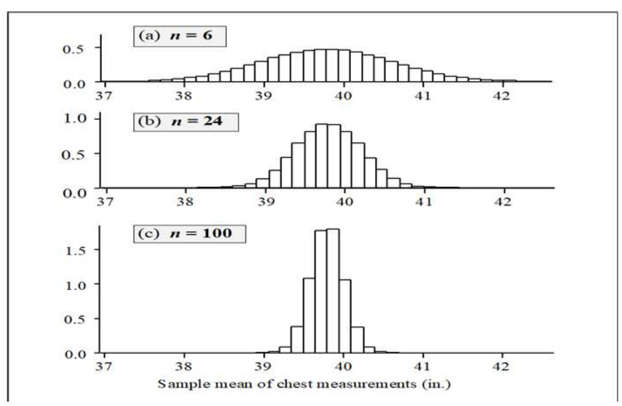


From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 1999.

12

Example

- With twelve samples, the histogram is noisy
- Instead of twelve samples of two different sizes, we take 100,000 samples of 3 different sizes
- In each case, the distribution of the sample means is approximately normal



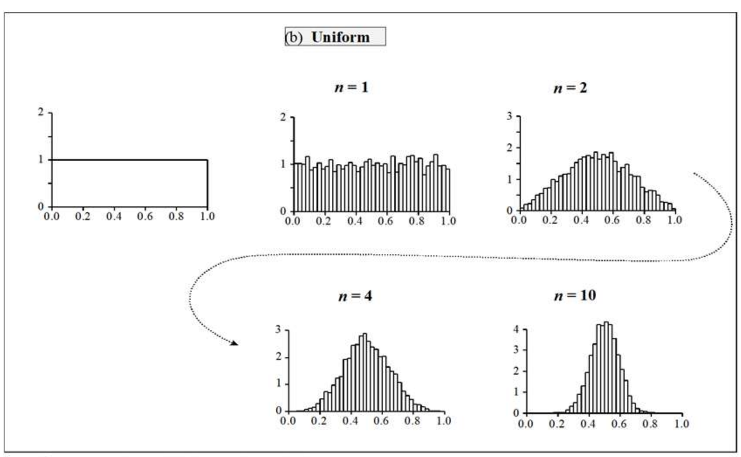
13

Why normal?

- We have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - What if the r.v. comes from a different distribution?
- Central limit theorem
  - For large n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
- How large is n?

14

Uniform distribution

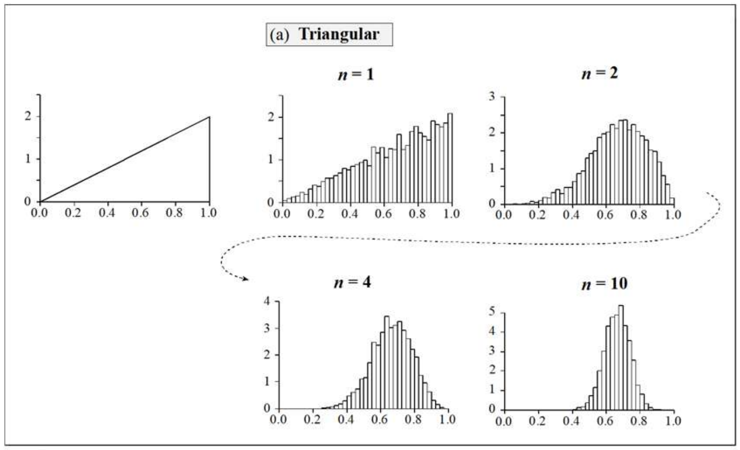


From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 2000.

5000 observations

15

Triangular distribution

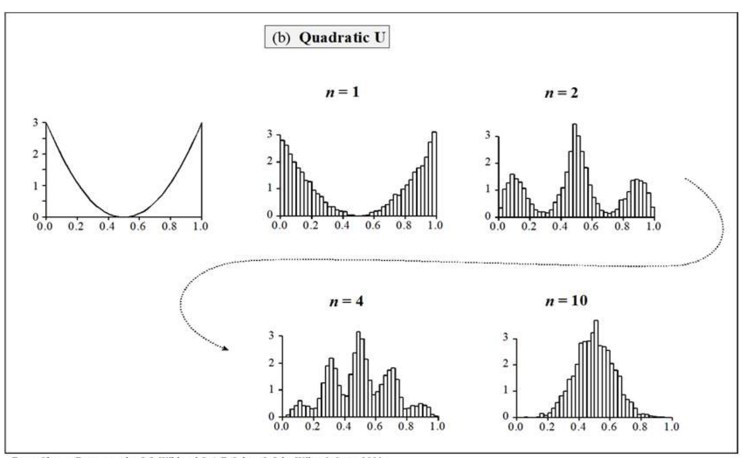


From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 2000.

5000 observations

16

Quadratic



From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 2000.

5000 observations

17